BRAC University
Department of Mathematics and Natural Sciences

Lecture on

Real Analysis II (MAT321)

Banach Fixed Point Theorem

Contraction Mapping, Banach Fixed Point Theorem, Picard-Lindelöf Theorem

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Conducted by

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Motivation of Fixed Point

Contraction Mapping

Contraction Mapping

Let (X, d) be a metric space. A function $: X \to X$ is called a *contraction mapping* if there exists a constant $0 < k < 1$ such that for all $x, y \in X$,

$$d(f(x), f(y)) \leq k \cdot d(x, y).$$

Example of Contraction Mapping

Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \frac{1}{2}x$. Show that f is a contraction mapping on $\mathbb{R}$ with the standard metric.

Another Example of Contraction Mapping

Let $X = [0, 1]$ and define $f : X \to X$ by $f(x) = \frac{1}{2}x + \frac{1}{4}$. Show that f is a contraction mapping on X.

Banach Fixed Point Theorem

Banach fixed point theorem

Let (X, d) be a complete metric space and let $f : X \to X$ be a contraction with constant $k \in [0, 1)$. Then:

(i) f has a unique fixed point $x^* \in X$;

(ii) for every initial point $x_0 \in X$, the iteration

$$x_{n+1} = f(x_n)$$

converges to x^*.

Derivative criterion for contraction on an interval

Let $f : [a, b] \to \mathbb{R}$ be a differentiable function. If there exists a constant $0 < k < 1$ such that $|f'(x)| \leq k$ for all $x \in [a, b]$, then f is a contraction mapping on $[a, b]$ with the standard metric.

Fixed point iteration method for solving equations

Solving ODEs using Banach fixed point theorem

Equivalent integral form of ODEs

Let $f : [a, b] \times \mathbb{R} \to \mathbb{R}$ be a continuous function. A function $y : [a, b] \to \mathbb{R}$ is a solution to the initial value problem

$$y' = f(x, y), \qquad y(a) = y_0$$

if and only if it satisfies the integral equation

$$y(x) = y_0 + \int_a^x f(t, y(t))\, dt.$$

Picard-Lindelof theorem and the ODE uniqueness theorem

Lipschitz condition

Let $f : [a, b] \times \mathbb{R} \to \mathbb{R}$ be a function. We say that f satisfies a *Lipschitz condition* with respect to the second variable if there exists a constant $L > 0$ such that for all $x \in [a, b]$ and all $u, v \in \mathbb{R}$,

$$|f(x, u) - f(x, v)| \leq L|u - v|.$$

Picard–Lindelöf / local existence and uniqueness

Suppose f is continuous on a rectangle

$$R = [t_0 - a, t_0 + a] \times [y_0 - b, y_0 + b]$$

and Lipschitz in y there with constant L. Let

$$M = \sup_{(t,y)\in R} |f(t,y)|, \qquad h = \min\left\{a, \frac{b}{M}, \frac{1}{L}\right\}$$

(with the convention $1/L = \infty$ if $L = 0$). Then the IVP

$$y' = f(t,y), \qquad y(t_0) = y_0$$

has a unique solution on $I = [t_0 - h, t_0 + h]$.

Picard iteration

Picard iteration for solving ODEs

Let $f : [a, b] \times \mathbb{R} \to \mathbb{R}$ be a continuous function. The Picard iteration is defined as follows:

$$y_0(x) = y_0, \qquad y_{n+1}(x) = y_0 + \int_a^x f(t, y_n(t))\, dt.$$

Thank You!

We'd love your questions and feedback.

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(Lectures, walkthroughs, and course updates)

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References

[1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.

[2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.